

Week 4 Linear Algebra

By

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General Vector Spaces

4.1 Real Vector Spaces

In this section we will extend the concept of a vector by using the basic properties of vectors in R^n as axioms, which if satisfied by a set of objects will guarantee that those objects behave like familiar vectors.

Vector Space Axioms

In this text scalars will be either real numbers or complex numbers. Vector spaces with real scalars will be called ***real vector spaces*** and those with complex scalars will be called ***complex vector spaces***. For now we will consider only real vector spaces.

Definition 1

Let V be an arbitrary nonempty set of objects for which two operations are defined: addition and multiplication by numbers called **scalars**. By **addition** we mean a rule for associating with each pair of objects $\mathbf{u}$ and $\mathbf{v}$ in V an object $\mathbf{u} + \mathbf{v}$, called the **sum** of $\mathbf{u}$ and $\mathbf{v}$; by **scalar multiplication** we mean a rule for associating with each scalar k and each object $\mathbf{u}$ in V an object $k\mathbf{u}$, called the **scalar multiple** of $\mathbf{u}$ by k . If the following axioms are satisfied by all objects $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V and all scalars k and m , then we call V a **vector space** and we call the objects in V **vectors**.

1. If $\mathbf{u}$ and $\mathbf{v}$ are objects in V , then $\mathbf{u} + \mathbf{v}$ is in V .
2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
4. There exists an object in V , called the **zero vector**, that is denoted by $\mathbf{0}$ and has the property that $\mathbf{0} + \mathbf{u} = \mathbf{u} + \mathbf{0} = \mathbf{u}$ for all $\mathbf{u}$ in V .
5. For each $\mathbf{u}$ in V , there is an object $-\mathbf{u}$ in V , called a **negative** of $\mathbf{u}$, such that $\mathbf{u} + (-\mathbf{u}) = (-\mathbf{u}) + \mathbf{u} = \mathbf{0}$.
6. If k is any scalar and $\mathbf{u}$ is any object in V , then $k\mathbf{u}$ is in V .
7. $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$
8. $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$
9. $k(m\mathbf{u}) = (km)(\mathbf{u})$
10. $1\mathbf{u} = \mathbf{u}$

EXAMPLE 1 | The Zero Vector Space

Let V consist of a single object, which we denote by $\mathbf{0}$, and define

$$\mathbf{0} + \mathbf{0} = \mathbf{0} \quad \text{and} \quad k\mathbf{0} = \mathbf{0}$$

for all scalars k . It is easy to check that all the vector space axioms are satisfied. We call this the *zero vector space*.

Our second example is one of the most important of all vector spaces—the familiar space R^n . It should not be surprising that the operations on R^n satisfy the vector space axioms because those axioms were based on known properties of operations on R^n .

EXAMPLE 2 | R^n Is a Vector Space

Let $V = R^n$, and define the vector space operations on V to be the usual operations of addition and scalar multiplication of n -tuples; that is,

$$\mathbf{u} + \mathbf{v} = (u_1, u_2, \dots, u_n) + (v_1, v_2, \dots, v_n) = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$$
$$k\mathbf{u} = (ku_1, ku_2, \dots, ku_n)$$

The set $V = R^n$ is closed under addition and scalar multiplication because the foregoing operations produce n -tuples as their end result, and these operations satisfy Axioms 2, 3, 4, 5, 7, 8, 9, and 10 by virtue of Theorem 3.1.1.

Historical Note



**Hermann Günther
Grassmann
(1809–1877)**

The notion of an “abstract vector space” evolved over many years and had many contributors. The idea crystallized with the work of the German mathematician H. G. Grassmann, who published a paper in 1862 in which he considered abstract systems of unspecified elements on which he defined formal operations of addition and scalar multiplication. Grassmann’s work was controversial, and others, including Augustin Cauchy (p. 136), laid reasonable claim to the idea.

[Image: © Sueddeutsche Zeitung Photo/The Image Works]

EXAMPLE 3 | The Vector Space of Infinite Sequences of Real Numbers

Let V consist of objects of the form

$$\mathbf{u} = (u_1, u_2, \dots, u_n, \dots)$$

in which $u_1, u_2, \dots, u_n, \dots$ is an infinite sequence of real numbers. We define two infinite sequences to be *equal* if their corresponding components are equal, and we define addition and scalar multiplication componentwise by

$$\begin{aligned}\mathbf{u} + \mathbf{v} &= (u_1, u_2, \dots, u_n, \dots) + (v_1, v_2, \dots, v_n, \dots) \\ &= (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n, \dots) \\ k\mathbf{u} &= (ku_1, ku_2, \dots, ku_n, \dots)\end{aligned}$$

In the exercises we ask you to confirm that V with these operations is a vector space. We will denote this vector space by the symbol R^∞ .

EXAMPLE 7 | A Set That Is Not a Vector Space

Let $V = R^2$ and define addition and scalar multiplication operations as follows: If $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$, then define

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2)$$

and if k is any real number, then define

$$k\mathbf{u} = (ku_1, 0)$$

For example, if $\mathbf{u} = (2, 4)$, $\mathbf{v} = (-3, 5)$, and $k = 7$, then

$$\mathbf{u} + \mathbf{v} = (-1, 9)$$

$$k\mathbf{u} = 7\mathbf{u} = (14, 0)$$

The addition operation is the standard one from R^2 , but the scalar multiplication is not. In the exercises we will ask you to show that the first nine vector space axioms are satisfied, but Axiom 10 fails to hold for certain vectors. For example, if $\mathbf{u} = (u_1, u_2)$ is such that $u_2 \neq 0$, then

$$1\mathbf{u} = 1(u_1, u_2) = (u_1, 0) \neq \mathbf{u}$$

Thus, V is not a vector space with the stated operations.

EXAMPLE 8 | An Unusual Vector Space

Let V be the set of positive real numbers, let $\mathbf{u} = u$ and $\mathbf{v} = v$ be any vectors (i.e., positive real numbers) in V , and let k be any scalar. Define the operations on V to be

$$u + v = uv \quad \text{[Vector addition is numerical multiplication.]}$$

$$ku = u^k \quad \text{[Scalar multiplication is numerical exponentiation.]}$$

Thus, for example, $1 + 1 = 1$ and $(2)(1) = 1^2 = 1$ —strange indeed, but nevertheless the set V with these operations satisfies the ten vector space axioms and hence is a vector space. We will confirm Axioms 4, 5, and 7, and leave the others as exercises.

- **Axiom 4**—The zero vector in this space is the number 1 (i.e., $\mathbf{0} = 1$) since

$$u + 1 = u \cdot 1 = u$$

- **Axiom 5**—The negative of a vector u is its reciprocal (i.e., $-u = 1/u$) since

$$u + \frac{1}{u} = u \left(\frac{1}{u} \right) = 1 (= \mathbf{0})$$

- **Axiom 7**— $k(u + v) = (uv)^k = u^k v^k = (ku) + (kv)$.

Theorem 4.1.1

Let V be a vector space, $\mathbf{u}$ a vector in V , and k a scalar; then:

(a) $0\mathbf{u} = \mathbf{0}$

(b) $k\mathbf{0} = \mathbf{0}$

(c) $(-1)\mathbf{u} = -\mathbf{u}$

(d) If $k\mathbf{u} = \mathbf{0}$, then $k = 0$ or $\mathbf{u} = \mathbf{0}$.

Exercise Set 4.1

1. Let V be the set of all ordered pairs of real numbers, and consider the following addition and scalar multiplication operations on $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2), \quad k\mathbf{u} = (0, ku_2)$$

- a. Compute $\mathbf{u} + \mathbf{v}$ and $k\mathbf{u}$ for $\mathbf{u} = (-1, 2)$, $\mathbf{v} = (3, 4)$, and $k = 3$.
 - b. In words, explain why V is closed under addition and scalar multiplication.
 - c. Since addition on V is the standard addition operation on $\mathbb{R}^2$, certain vector space axioms hold for V because they are known to hold for $\mathbb{R}^2$. Which axioms are they?
 - d. Show that Axioms 7, 8, and 9 hold.
 - e. Show that Axiom 10 fails and hence that V is not a vector space under the given operations.
2. Let V be the set of all ordered pairs of real numbers, and consider the following addition and scalar multiplication operations on $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1 + 1, u_2 + v_2 + 1), \quad k\mathbf{u} = (ku_1, ku_2)$$

- a. Compute $\mathbf{u} + \mathbf{v}$ and $k\mathbf{u}$ for $\mathbf{u} = (0, 4)$, $\mathbf{v} = (1, -3)$, and $k = 2$.
- b. Show that $(0, 0) \neq \mathbf{0}$.
- c. Show that $(-1, -1) = \mathbf{0}$.

4.2 Subspaces

Definition 1

A subset W of a vector space V is called a **subspace** of V if W is itself a vector space under the addition and scalar multiplication defined on V .

Theorem 4.2.1

Subspace Test

If W is a nonempty set of vectors in a vector space V , then W is a subspace of V if and only if the following conditions are satisfied.

- (a) If $\mathbf{u}$ and $\mathbf{v}$ are vectors in W , then $\mathbf{u} + \mathbf{v}$ is in W .
- (b) If k is a scalar and $\mathbf{u}$ is a vector in W , then $k\mathbf{u}$ is in W .

The Subspace Test states that W is a subspace of V if and only if it is closed under addition and scalar multiplication.

EXAMPLE 1 | The Zero Subspace

If V is any vector space, and if $W = \{\mathbf{0}\}$ is the subset of V that consists of the zero vector only, then W is closed under addition and scalar multiplication since

$$\mathbf{0} + \mathbf{0} = \mathbf{0} \quad \text{and} \quad k\mathbf{0} = \mathbf{0}$$

for any scalar k . We call W the **zero subspace** of V .

Thank You